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FUTURE TECH SCHOOL PROGRAM

Faculty Name :

Course Duration : 5 Months

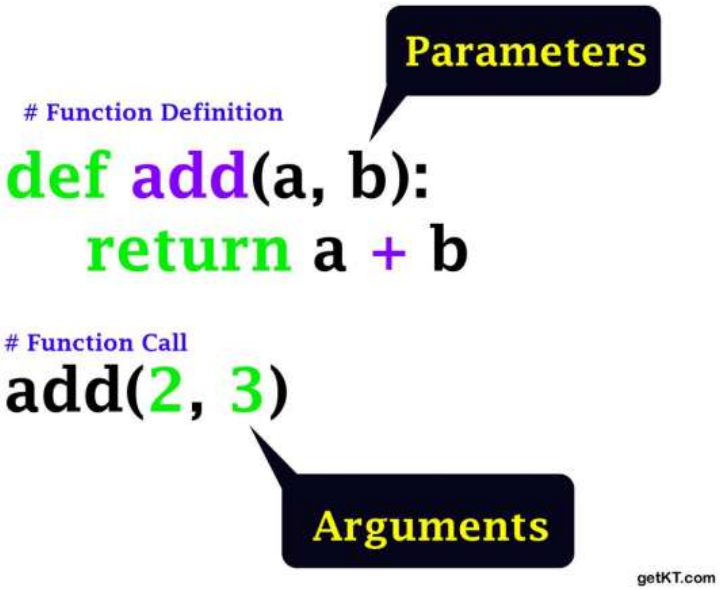
Classes : Offline & Online

Daily Teaching : 2 Hours

Total Classes : 100

Level 2 – Young Technologist (7th–8th Standard)

Lesson 26: Function Parameters in Python



Learning Objectives

By the end of this lesson, students will be able to:

- Understand what **Function Parameters** are.
- Learn the difference between **Parameters** and **Arguments**.

- Create functions that accept one or more parameters.
- Pass different values to functions.
- Use parameters to solve real-life problems.
- Write simple Python programs using function parameters.

What are Function Parameters?

A **Function Parameter** is a **variable** written inside the parentheses of a function. It receives information (input) when the function is called.

Think of a function as a **machine**. The machine needs something to work on. That "something" is called a **parameter**.

Real-Life Example

Imagine a **juice blender**:

- Put an **apple** → Apple juice.
- Put a **mango** → Mango juice.
- Put a **banana** → Banana shake.

The **blender** is the **function**, and the **fruit** you put into it is the **parameter**.

Real-Life Example of Function Parameters



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Why Do We Use Parameters?

Parameters help us:

- Reuse the same function with different values.
- Avoid writing the same code again.
- Make programs flexible.
- Keep code clean and organized.

Function with One Parameter

Syntax

```
def function_name(parameter):
    # Code


function_name(value)
```

Example 1

```
def greet(name):
    print("Hello", name)


greet("Rahul")
```

Output

```
Hello Rahul
```

Example 2

```
def welcome(student):
```

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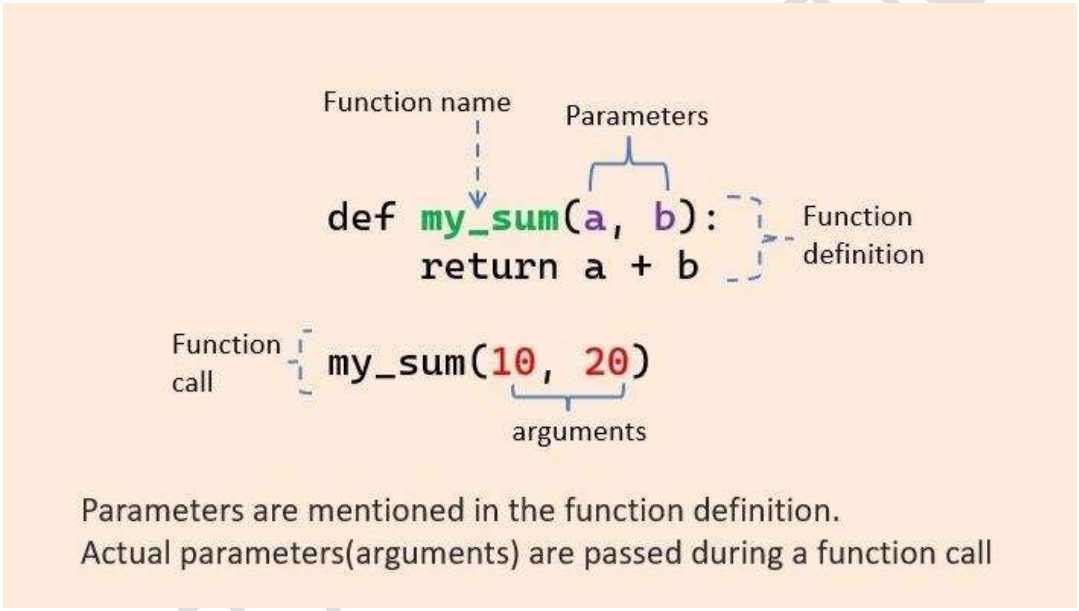
```
print("Welcome", student)

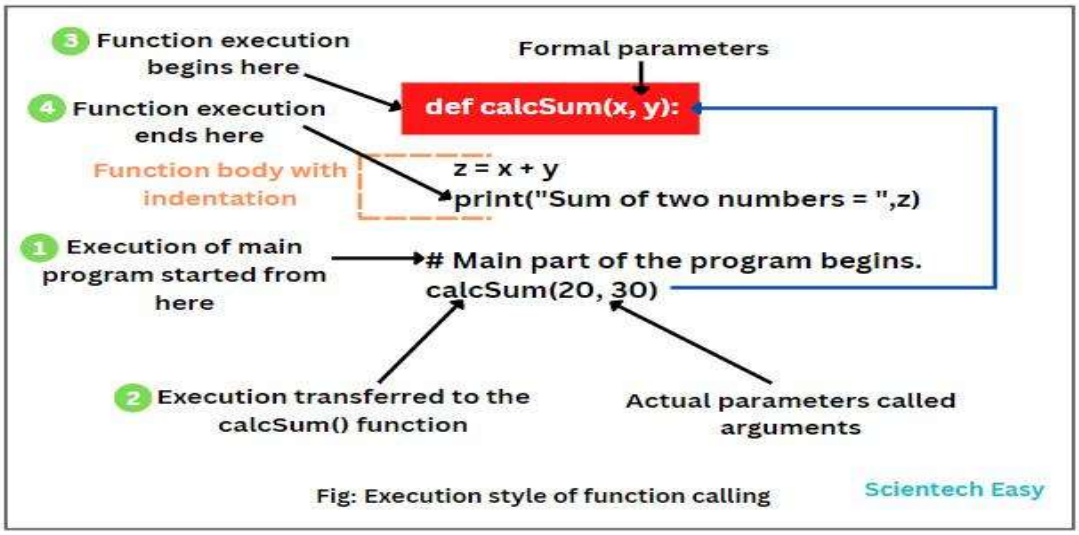

welcome("Anjali")
```

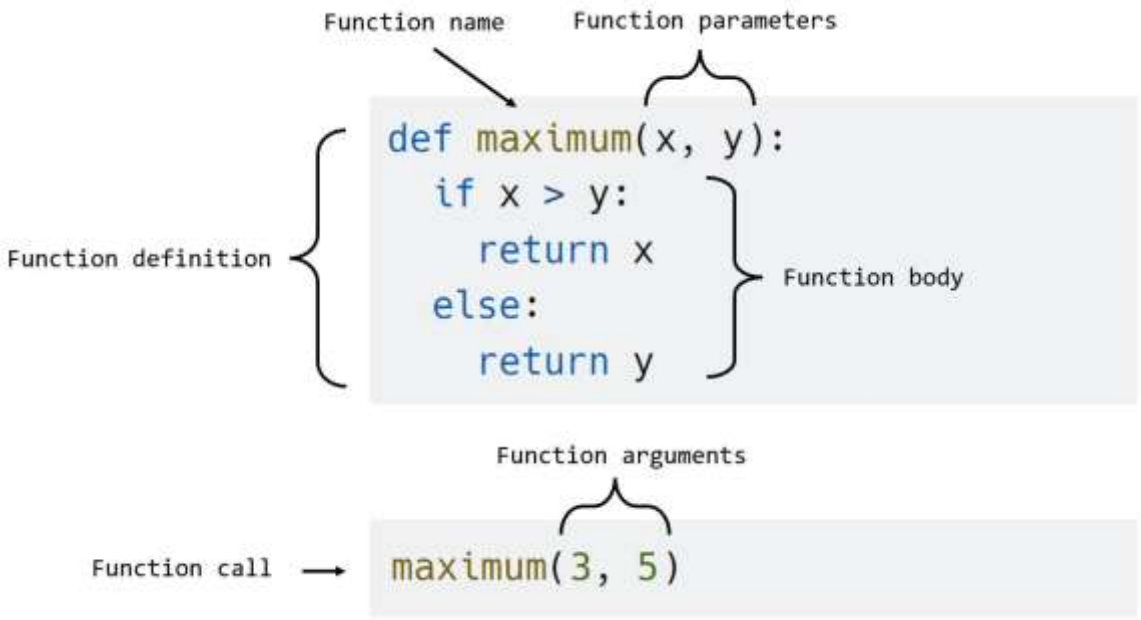
Output

Welcome Anjali

Parameter Passing







Function with Two Parameters

Syntax

```
def function_name(parameter1, parameter2):
    # Code
```

Example 3

```
def add(a, b):
    print("Sum =", a + b)

add(10, 20)
```

Output

Sum = 30

Example 4

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```
def student(name, age):
    print("Name:", name)
    print("Age:", age)

student("Asha", 13)
```

Output

```
Name: Asha
Age: 13
```

Function with Three Parameters

```
def school(name, grade, city):
    print("Name:", name)
    print("Grade:", grade)
    print("City:", city)

school("Rahul", "7th", "Hyderabad")
```

Output

```
Name: Rahul
Grade: 7th
City: Hyderabad
```

Parameters vs Arguments

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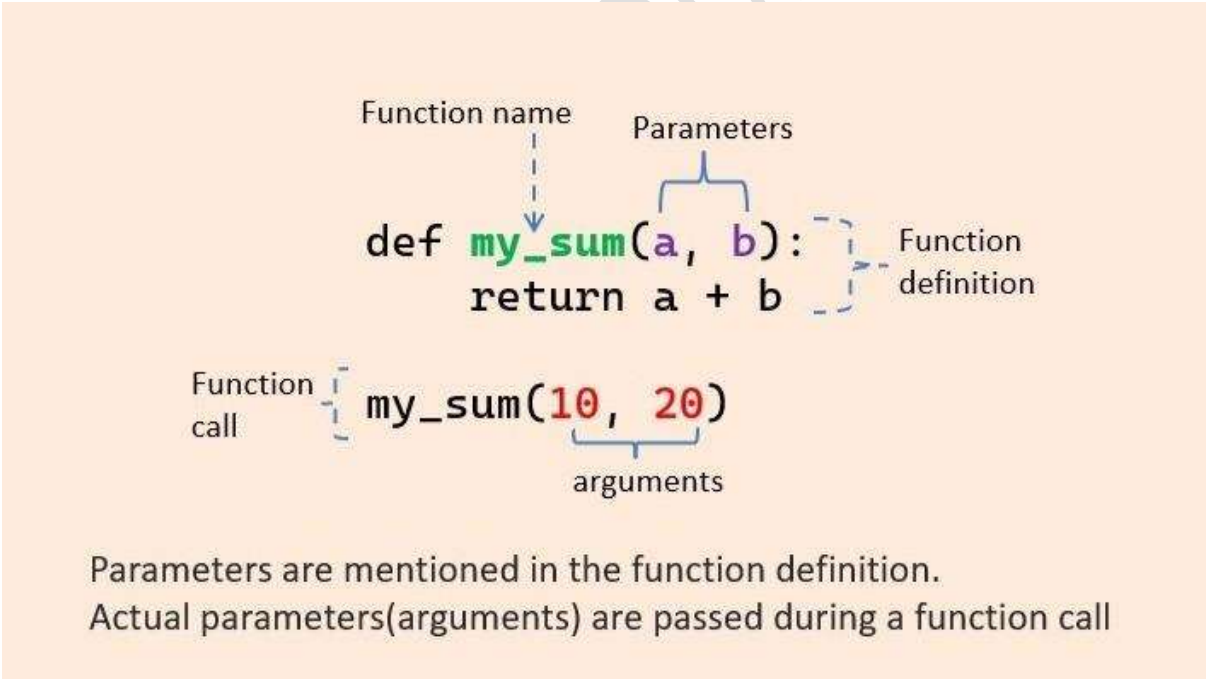
Parameter	Argument
Variable in a function definition	Actual value passed to the function
Written inside def	Written when calling the function

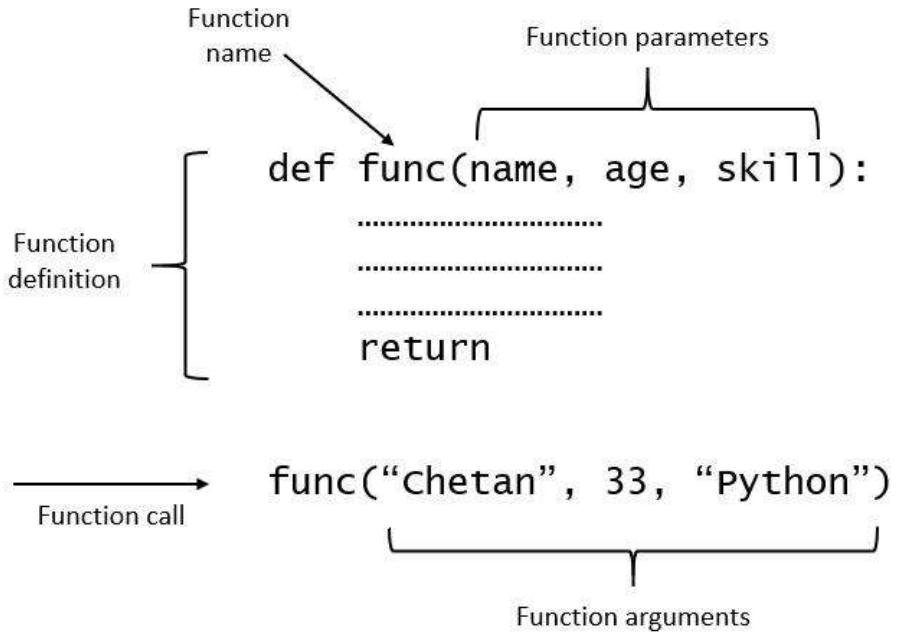
Example

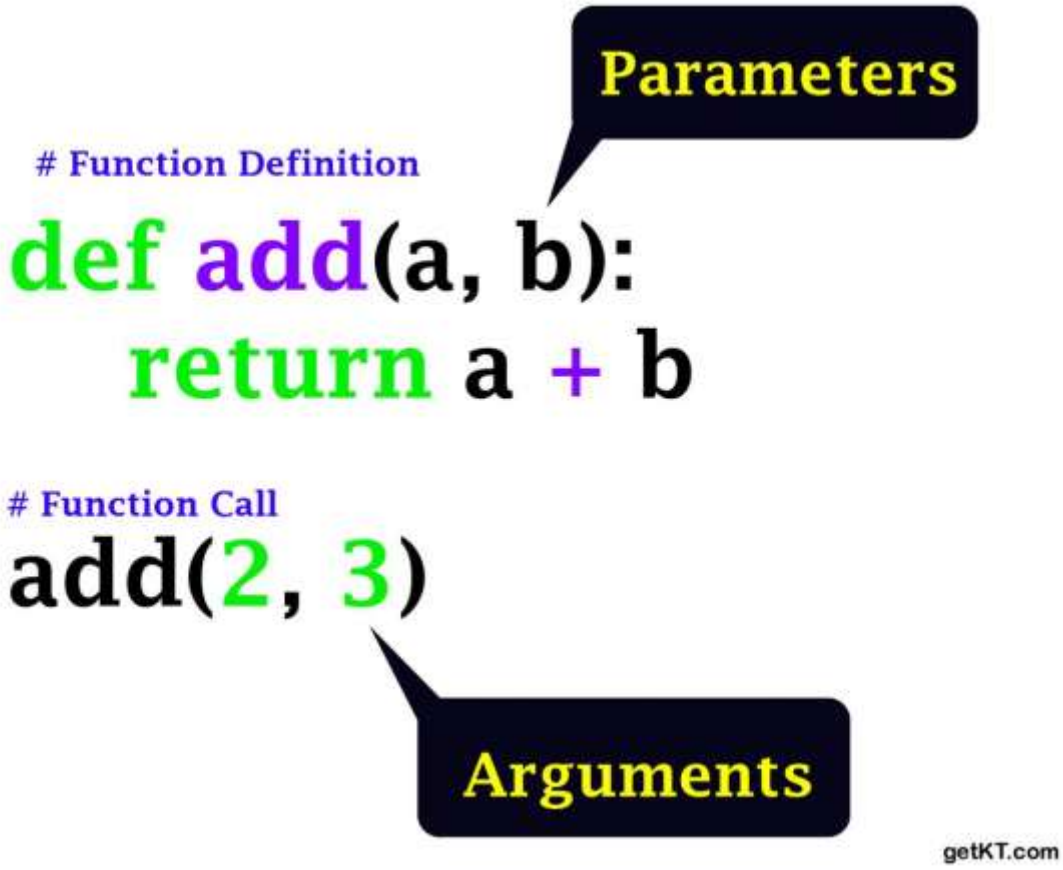
```
def greet(name):     # name → Parameter
    print(name)

greet("Rahul")       # Rahul → Argument
```

Parameters vs Arguments







Function Returning a Value

```
def multiply(a, b):
    return a * b

result = multiply(5, 6)

print(result)
```

Output

30

Example 6

```
def square(number):
    return number * number

print(square(8))
```

Output

64

Example 7

```
def area(length, breadth):
    return length * breadth

print(area(12, 5))
```

Output

60

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Built-in Functions with Parameters

Python also has built-in functions that accept parameters.

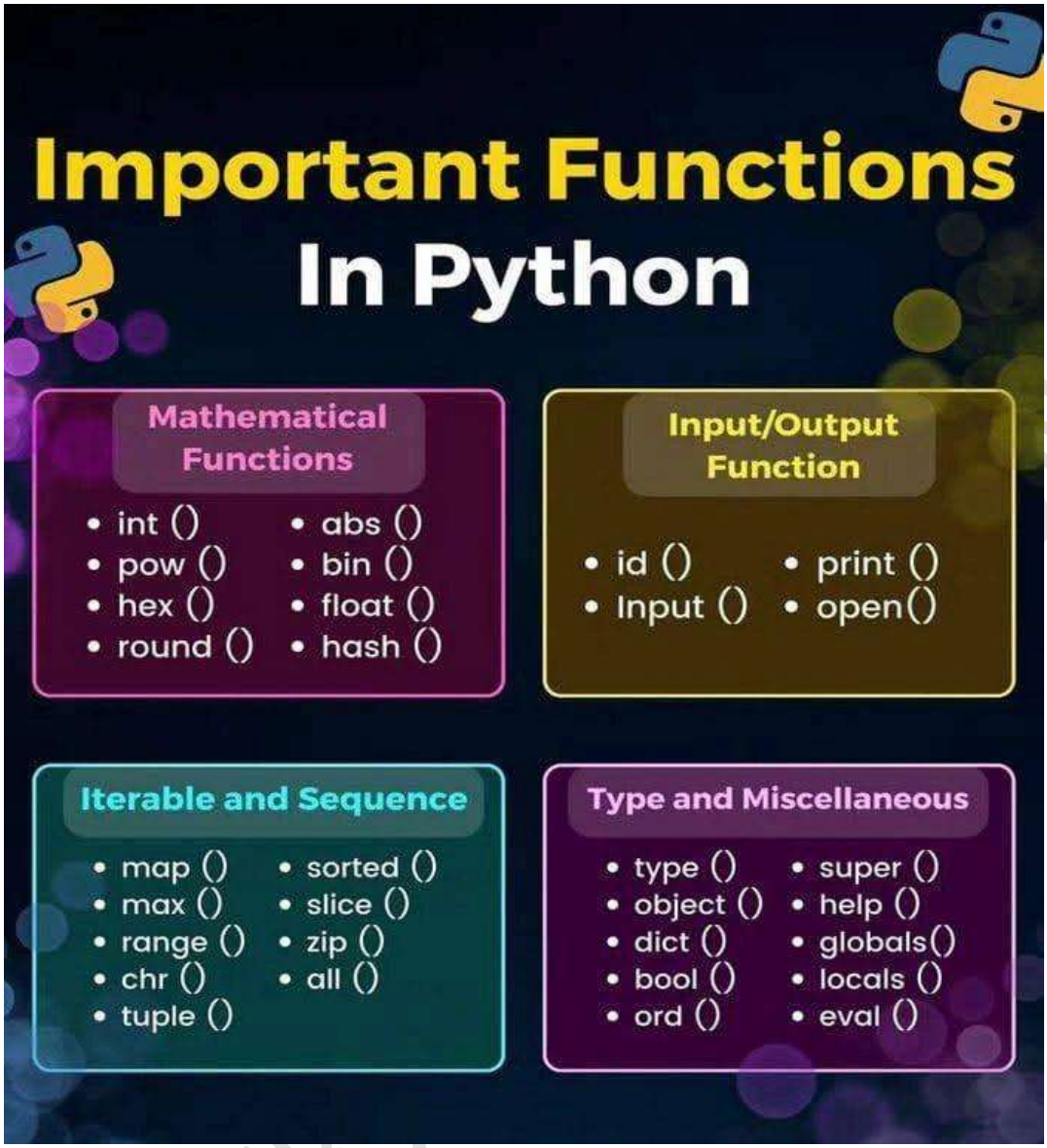
print("Python")

len("Computer")

int("25")

Built-in Functions





PYTHON BUILT IN FUNCTIONS

Abs()-Returns the absolute value of a number

For Example, x = abs(-7.25)

All()-Returns True if all items in an iterable object are true

For Example, mylist = [True, True, True]
x = all(mylist)

Bin()-Returns the binary version of a number

For Example, x = bin(36)

Bool()-Returns the boolean value of the specified object

For Example, x = bool(1)

Practical Session

Practical Activity 1

Greeting Function

```
def greet(name):
    print("Hello", name)


greet(input("Enter Your Name: "))
```

Practical Activity 2

Addition Function

```
def add(a, b):
    print("Sum =", a + b)


add(12, 18)
```

Practical Activity 3

Area of Rectangle

```
def area(length, breadth):
    print("Area =", length * breadth)


area(8, 5)
```

Practical Activity 4

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Student Information

```
def student(name, age):
    print("Name:", name)
    print("Age:", age)

student("Kiran", 12)
```

Practical Activity 5

Multiplication Function

```
def multiply(a, b):
    print("Product =", a * b)

multiply(7, 9)
```

Practical Activity 6

Square Function

```
def square(number):
    print("Square =", number * number)

square(11)
```

Practical Activity 7

Even or Odd

```
def even(number):
```

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```
    if number % 2 == 0:
        print("Even")
    else:
        print("Odd")

even(17)
```

Practical Activity 8

Multiplication Table

```
def table(number):
    for i in range(1, 11):
        print(number, "x", i, "=", number * i)

table(5)
```

Classroom Coding Activity



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Activity 1

Create a function to print your favorite subject using a parameter.

Activity 2

Create a function to calculate the sum of two numbers.

Activity 3

Create a function that displays student details.

Activity 4

Create a function that prints the multiplication table of any number.

Real-Life Applications

Situation	Function Parameter Use
Calculator	Numbers are passed as parameters
ATM	Account number and PIN
School App	Student name and roll number
Shopping App	Product name and quantity
Game	Player name and score
Banking	Amount and account details

Common Mistakes

❌ **Forgetting to Pass an Argument**

Wrong

```
def greet(name):
    print(name)
```

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```
greet()
```

❌ Error: Missing required argument.

✅ **Correct**

```
def greet(name):
    print(name)

greet("Rahul")
```

❌ **Wrong Number of Arguments**

Wrong

```
def add(a, b):
    print(a + b)

add(10)
```

✅ **Correct**

```
add(10, 20)
```

❌ **Incorrect Indentation**

Wrong

```
def greet(name):
print(name)
```

✅ **Correct**

```
def greet(name):
```

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```
print(name)
```

Best Practices

- ✓ Use meaningful parameter names (name, age, marks).
- ✓ Pass the correct number of arguments.
- ✓ Keep functions short and focused.
- ✓ Reuse functions instead of repeating code.
- ✓ Test with different inputs.

Keywords

- **Function** – A reusable block of code.
- **Parameter** – Variable in the function definition.
- **Argument** – Value passed to the function.
- **Return** – Sends a value back from a function.
- **Function Call** – Executes a function.

Lesson Summary

- A **parameter** is a variable used in a function definition.
- An **argument** is the actual value passed to the function.
- Functions can have one, two, or many parameters.
- Parameters make functions flexible and reusable.
- The return statement sends a result back to the program.

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- Using parameters helps reduce repeated code and makes programs easier to maintain.

📝 Quiz

Multiple Choice Questions

1. What is a parameter?

A) A loop

B) A variable in a function definition

C) A list

D) An operator

✅ **Answer: B**

---

2. What is an argument?

A) The value passed to a function

B) A loop

C) A variable

D) A condition

✅ **Answer: A**

---

3. Which keyword is used to define a function?

A) if

B) for

C) def

D) while

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✅ **Answer: C**

---

4. Which keyword returns a value from a function?

A) break

B) continue

C) return

D) input

✅ **Answer: C**

---

5. What will this code print?

```
def greet(name):
    print("Hello", name)

greet("Anjali")
```

A) Hello

B) Anjali

C) Hello Anjali

D) Error

✅ **Answer: C**

🏠 **Homework**

1. Write a function that prints your name using a parameter.
2. Write a function to add two numbers.

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3. Write a function to calculate the area of a rectangle.
4. Write a function to display student information (name and age).
5. Write a function to return the square of a number.
6. Write a function to check whether a number is even or odd.
7. Write a function to display the multiplication table of any number.

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